Zookeeper (manager)

ZooKeeper is a centralized service for maintaining configuration information, naming, providing distributed synchronization, and providing group services. All of these kinds of services are used in some form or another by distributed applications. Each time they are implemented there is a lot of work that goes into fixing the bugs and race conditions that are inevitable. Because of the difficulty of implementing these kinds of services, applications initially usually skimp on them, which makes them brittle in the presence of change and difficult to manage. Even when done correctly, different implementations of these services lead to management complexity when the applications are deployed.

```
sistla_keerthana@Sistla: ~/kaf
[2026-04-03 14:42:26,916] INFO  (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,916] INFO   ______                   _                                          (org.apache.zookeeper.server.Zoo
KeeperServer)
[2026-04-03 14:42:26,916] INFO  |___  /                  | |                                         (org.apache.zookeeper.server.Zoo
KeeperServer)
[2026-04-03 14:42:26,916] INFO     / /    ___     ___   | | __   ___    ___   _ __     ___   _ __    (org.apache.zookeeper.server.Zoo
KeeperServer)
[2026-04-03 14:42:26,916] INFO    / /    / _ \   / _ \  | |/ /  / _ \  / _ \ | '_ \   / _ \ | '__| (org.apache.zookeeper.server.ZooKe
eperServer)
[2026-04-03 14:42:26,916] INFO   / /__  | (_) | | (_) | |   <  |  __/ |  __/ | |_) | |  __/ | |     (org.apache.zookeeper.server.ZooK
eeperServer)
[2026-04-03 14:42:26,917] INFO  /_____|  \___/   \___/  |_|\_\  \___|  \___| | .__/   \___| |_| (org.apache.zookeeper.server.ZooKeepe
rServer)
[2026-04-03 14:42:26,917] INFO                                               | |                      (org.apache.zookeeper.server.Zo
oKeeperServer)
[2026-04-03 14:42:26,917] INFO                                               |_|                      (org.apache.zookeeper.server.Zo
oKeeperServer)
[2026-04-03 14:42:26,917] INFO  (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,921] INFO Server environment:zookeeper.version=3.8.3-6ad6d364c7c0bcf0de452d54ebefa3058098ab56, built on 2023-10-
05 10:34 UTC (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,922] INFO Server environment:host.name=Sistla.localdomain (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,922] INFO Server environment:java.version=1.8.0_482 (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,922] INFO Server environment:java.vendor=Private Build (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,922] INFO Server environment:java.home=/usr/lib/jvm/java-8-openjdk-amd64/jre (org.apache.zookeeper.server.ZooKee
perServer)
[2026-04-03 14:42:26,922] INFO Server environment:java.class.path=/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/activation-1.1.
1.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/aopalliance-repackaged-2.6.1.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bi
n/../libs/argparse4j-0.7.0.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/audience-annotations-0.12.0.jar:/home/sistla_keert
hana/kafka_2.13-3.7.0/bin/../libs/caffeine-2.9.3.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/checker-qual-3.19.0.jar:/hom
e/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/commons-beanutils-1.9.4.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/commo
ns-cli-1.4.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/commons-collections-3.2.2.jar:/home/sistla_keerthana/kafka_2.13-3.
7.0/bin/../libs/commons-digester-2.1.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/commons-io-2.11.0.jar:/home/sistla_keert
hana/kafka_2.13-3.7.0/bin/../libs/commons-lang3-3.8.1.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/commons-logging-1.2.jar
:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/commons-validator-1.7.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/co
nnect-api-3.7.0.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/connect-basic-auth-extension-3.7.0.jar:/home/sistla_keerthana
/kafka_2.13-3.7.0/bin/../libs/connect-json-3.7.0.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/connect-mirror-3.7.0.jar:/ho
me/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/connect-mirror-client-3.7.0.jar:/home/sistla_keerthana/kafka_2.13-3.7.0/bin/../libs/
```

```
sistla_keerthana@Sistla: ~/kaf
[2026-04-03 14:42:26,935] INFO zookeeper.pathStats.delay = 5 (org.apache.zookeeper.server.util.RequestPathMetricsCollector)
[2026-04-03 14:42:26,935] INFO zookeeper.pathStats.enabled = false (org.apache.zookeeper.server.util.RequestPathMetricsCollector)
[2026-04-03 14:42:26,939] INFO The max bytes for all large requests are set to 104857600 (org.apache.zookeeper.server.ZooKeeperServer
)
[2026-04-03 14:42:26,940] INFO The large request threshold is set to -1 (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,940] INFO zookeeper.enforce.auth.enabled = false (org.apache.zookeeper.server.AuthenticationHelper)
[2026-04-03 14:42:26,941] INFO zookeeper.enforce.auth.schemes = [] (org.apache.zookeeper.server.AuthenticationHelper)
[2026-04-03 14:42:26,941] INFO Created server with tickTime 3000 ms minSessionTimeout 6000 ms maxSessionTimeout 60000 ms clientPortLi
stenBacklog -1 datadir /tmp/zookeeper/version-2 snapdir /tmp/zookeeper/version-2 (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:26,957] INFO Using org.apache.zookeeper.server.NIOServerCnxnFactory as server connection factory (org.apache.zookee
per.server.ServerCnxnFactory)
[2026-04-03 14:42:26,962] WARN maxCnxns is not configured, using default value 0. (org.apache.zookeeper.server.ServerCnxnFactory)
[2026-04-03 14:42:26,967] INFO Configuring NIO connection handler with 10s sessionless connection timeout, 2 selector thread(s), 24 w
orker threads, and 64 kB direct buffers. (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2026-04-03 14:42:26,985] INFO binding to port 0.0.0.0/0.0.0.0:2181 (org.apache.zookeeper.server.NIOServerCnxnFactory)
[2026-04-03 14:42:27,096] INFO Using org.apache.zookeeper.server.watch.WatchManager as watch manager (org.apache.zookeeper.server.wat
ch.WatchManagerFactory)
[2026-04-03 14:42:27,096] INFO Using org.apache.zookeeper.server.watch.WatchManager as watch manager (org.apache.zookeeper.server.wat
ch.WatchManagerFactory)
[2026-04-03 14:42:27,097] INFO zookeeper.snapshotSizeFactor = 0.33 (org.apache.zookeeper.server.ZKDatabase)
[2026-04-03 14:42:27,097] INFO zookeeper.commitLogCount=500 (org.apache.zookeeper.server.ZKDatabase)
[2026-04-03 14:42:27,111] INFO zookeeper.snapshot.compression.method = CHECKED (org.apache.zookeeper.server.persistence.SnapStream)
[2026-04-03 14:42:27,111] INFO Snapshotting: 0x0 to /tmp/zookeeper/version-2/snapshot.0 (org.apache.zookeeper.server.persistence.File
TxnSnapLog)
[2026-04-03 14:42:27,117] INFO Snapshot loaded in 20 ms, highest zxid is 0x0, digest is 1371985504 (org.apache.zookeeper.server.ZKDat
abase)
[2026-04-03 14:42:27,120] INFO Snapshotting: 0x0 to /tmp/zookeeper/version-2/snapshot.0 (org.apache.zookeeper.server.persistence.File
TxnSnapLog)
[2026-04-03 14:42:27,122] INFO Snapshot taken in 1 ms (org.apache.zookeeper.server.ZooKeeperServer)
[2026-04-03 14:42:27,139] INFO PrepRequestProcessor (sid:0) started, reconfigEnabled=false (org.apache.zookeeper.server.PrepRequestPr
ocessor)
[2026-04-03 14:42:27,140] INFO zookeeper.request_throttler.shutdownTimeout = 10000 ms (org.apache.zookeeper.server.RequestThrottler)
[2026-04-03 14:42:27,167] INFO Using checkIntervalMs=60000 maxPerMinute=10000 maxNeverUsedIntervalMs=0 (org.apache.zookeeper.server.C
ontainerManager)
[2026-04-03 14:42:27,168] INFO ZooKeeper audit is disabled. (org.apache.zookeeper.audit.ZKAuditProvider)
[2026-04-03 14:42:45,102] INFO Creating new log file: log.1 (org.apache.zookeeper.server.persistence.FileTxnLog)
```

Kafka (server)

Apache Kafka is an open-source distributed event streaming platform used by thousands of companies for high-performance data pipelines, streaming analytics, data integration, and mission-critical applications.

```
[2026-04-03 14:42:45,119] INFO Session establishment complete on server localhost/127.0.0.1:2181, session id = 0x100000d9c8b0000, neg
otiated timeout = 18000 (org.apache.zookeeper.ClientCnxn)
[2026-04-03 14:42:45,124] INFO [ZooKeeperClient Kafka server] Connected. (kafka.zookeeper.ZooKeeperClient)
[2026-04-03 14:42:45,646] INFO Cluster ID = 0UYj9Lr_TNObHA18v_bE8g (kafka.server.KafkaServer)
[2026-04-03 14:42:45,737] INFO KafkaConfig values:
        advertised.listeners = null
        alter.config.policy.class.name = null
        alter.log.dirs.replication.quota.window.num = 11
        alter.log.dirs.replication.quota.window.size.seconds = 1
        authorizer.class.name =
        auto.create.topics.enable = true
        auto.include.jmx.reporter = true
        auto.leader.rebalance.enable = true
        background.threads = 10
        broker.heartbeat.interval.ms = 2000
        broker.id = 0
        broker.id.generation.enable = true
        broker.rack = null
        broker.session.timeout.ms = 9000
        client.quota.callback.class = null
        compression.type = producer
        connection.failed.authentication.delay.ms = 100
        connections.max.idle.ms = 600000
        connections.max.reauth.ms = 0
        control.plane.listener.name = null
        controlled.shutdown.enable = true
        controlled.shutdown.max.retries = 3
        controlled.shutdown.retry.backoff.ms = 5000
        controller.listener.names = null
        controller.quorum.append.linger.ms = 25
        controller.quorum.election.backoff.max.ms = 1000
        controller.quorum.election.timeout.ms = 1000
        controller.quorum.fetch.timeout.ms = 2000
        controller.quorum.request.timeout.ms = 2000
        controller.quorum.retry.backoff.ms = 20
        controller.quorum.voters = []
        controller.quota.window.num = 11
```

```
sistla_keerthana@Sistla: ~
[2026-04-03 14:55:26,808] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
27 in 19 milliseconds for epoch 0, of which 19 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,808] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
42 in 18 milliseconds for epoch 0, of which 18 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,808] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
12 in 18 milliseconds for epoch 0, of which 18 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,808] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
21 in 18 milliseconds for epoch 0, of which 18 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,809] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
36 in 19 milliseconds for epoch 0, of which 19 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,809] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
6 in 19 milliseconds for epoch 0, of which 19 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager)
[2026-04-03 14:55:26,809] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
43 in 19 milliseconds for epoch 0, of which 19 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,809] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
13 in 19 milliseconds for epoch 0, of which 19 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:26,810] INFO [GroupMetadataManager brokerId=0] Finished loading offsets and group metadata from __consumer_offsets-
28 in 20 milliseconds for epoch 0, of which 19 milliseconds was spent in the scheduler. (kafka.coordinator.group.GroupMetadataManager
)
[2026-04-03 14:55:27,453] INFO [GroupCoordinator 0]: Dynamic member with unknown member id joins group console-consumer-84854 in Empt
y state. Created a new member id console-consumer-f983f7eb-e7fb-4792-b19a-94d4b0b644ac and request the member to rejoin with this id.
 (kafka.coordinator.group.GroupCoordinator)
[2026-04-03 14:55:27,479] INFO [GroupCoordinator 0]: Preparing to rebalance group console-consumer-84854 in state PreparingRebalance
with old generation 0 (__consumer_offsets-20) (reason: Adding new member console-consumer-f983f7eb-e7fb-4792-b19a-94d4b0b644ac with g
roup instance id None; client reason: need to re-join with the given member-id: console-consumer-f983f7eb-e7fb-4792-b19a-94d4b0b644ac
) (kafka.coordinator.group.GroupCoordinator)
[2026-04-03 14:55:27,499] INFO [GroupCoordinator 0]: Stabilized group console-consumer-84854 generation 1 (__consumer_offsets-20) wit
h 1 members (kafka.coordinator.group.GroupCoordinator)
[2026-04-03 14:55:27,535] INFO [GroupCoordinator 0]: Assignment received from leader console-consumer-f983f7eb-e7fb-4792-b19a-94d4b0b
644ac for group console-consumer-84854 for generation 1. The group has 1 members, 0 of which are static. (kafka.coordinator.group.Gro
```

Producer (input)

Streaming input from command line interface

-> Creating Topics and Partitions

```
sistla_keerthana@Sistla: ~/kai
sistla_keerthana@Sistla:~$ cd ~/kafka_2.13-3.7.0

bin/kafka-topics.sh --create \
--topic stock-data \
--bootstrap-server localhost:9092 \
--partitions 3 \
--replication-factor 1
Created topic stock-data.
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ bin/kafka-topics.sh --describe \
--topic stock-data \
--bootstrap-server localhost:9092
Topic: stock-data       TopicId: mqOUgKd8RJSMJ5hqMVXX3g PartitionCount: 3       ReplicationFactor: 1    Configs:
        Topic: stock-data       Partition: 0    Leader: 0       Replicas: 0     Isr: 0
        Topic: stock-data       Partition: 1    Leader: 0       Replicas: 0     Isr: 0
        Topic: stock-data       Partition: 2    Leader: 0       Replicas: 0     Isr: 0
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ bin/kafka-console-producer.sh \
--topic stock-data \
--bootstrap-server localhost:9092
>AAPL,250
>TSLA,300
>^Csistla_keerthana@Sistla:~/kafka_2.13-3.7.0$
```

Consumer

```
sistla_keerthana@Sistla:~$ cd ~/kafka_2.13-3.7.0
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ bin/kafka-console-consumer.sh \
--topic stock-data \
--from-beginning \
--bootstrap-server localhost:9092
AAPL,250
TSLA,300
```

Creating Files

```
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ cd ~
sistla_keerthana@Sistla:~$ mkdir kafka_project
cd kafka_project
sistla_keerthana@Sistla:~/kafka_project$ touch consumer_ops.py
touch producer_api.py
touch consumer_advanced.py
touch consumer_mongo.py
touch spark_consumer.py
sistla_keerthana@Sistla:~/kafka_project$ nano consumer_ops.py
sistla_keerthana@Sistla:~/kafka_project$ nano producer_api.py
sistla_keerthana@Sistla:~/kafka_project$ nano consumer_advanced.py
sistla_keerthana@Sistla:~/kafka_project$ nano consumer_mongo.py
sistla_keerthana@Sistla:~/kafka_project$ nano spark_consumer.py
```

Installing Requirements

```
sistla_keerthana@Sistla: ~/kai
sistla_keerthana@Sistla:~/kafka_project$ sudo apt install python3-pip
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  build-essential bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg-dev fakeroot g++ g++-13
  g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu
  javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libaom3 libasan8 libatomic1
  libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libde265-0 libdpkg-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libgcc-13-dev libgd3 libgomp1 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1
  libhwasan0 libisl23 libitm1 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0 libmpc3 libpython3-dev libpython3.12-dev
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libquadmath0 libstdc++-13-dev libtsan2 libubsan1 linux-libc-dev
  lto-disabled-list make manpages-dev python3-dev python3-wheel python3.12 python3.12-dev python3.12-minimal rpcsvc-proto
  zlib1g-dev
Suggested packages:
  bzip2-doc cpp-doc gcc-13-locales cpp-13-doc debian-keyring g++-multilib g++-13-multilib gcc-13-doc gcc-multilib autoconf automake
  libtool flex bison gdb gcc-doc gcc-13-multilib gdb-x86-64-linux-gnu apache2 | lighttpd | httpd glibc-doc bzr libgd-tools
  libheif-plugin-x265 libheif-plugin-ffmpegdec libheif-plugin-jpegdec libheif-plugin-jpegenc libheif-plugin-j2kdec
  libheif-plugin-j2kenc libheif-plugin-rav1e libheif-plugin-svtenc libstdc++-13-doc make-doc python3.12-venv python3.12-doc
  binfmt-support
```

Creating Virtual Environment

```
sistla_keerthana@Sistla:~/kafka_project$ python3 -m venv venv
The virtual environment was not created successfully because ensurepip is not
available.  On Debian/Ubuntu systems, you need to install the python3-venv
package using the following command.

    apt install python3.12-venv

You may need to use sudo with that command.  After installing the python3-venv
package, recreate your virtual environment.

Failing command: /home/sistla_keerthana/kafka_project/venv/bin/python3

sistla_keerthana@Sistla:~/kafka_project$ sudo apt install python3.12-venv
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pip-whl python3-setuptools-whl
The following NEW packages will be installed:
  python3-pip-whl python3-setuptools-whl python3.12-venv
0 upgraded, 3 newly installed, 0 to remove and 63 not upgraded.
Need to get 2429 kB of archives.
After this operation, 2777 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-pip-whl all 24.0+dfsg-1ubuntu1.3 [1707 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-setuptools-whl all 68.1.2-2ubuntu1.2 [716 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3.12-venv amd64 3.12.3-1ubuntu0.12 [5666 B]
Fetched 2429 kB in 2s (1069 kB/s)
Selecting previously unselected package python3-pip-whl.
(Reading database ... 52859 files and directories currently installed.)
Preparing to unpack .../python3-pip-whl_24.0+dfsg-1ubuntu1.3_all.deb ...
Unpacking python3-pip-whl (24.0+dfsg-1ubuntu1.3) ...
Selecting previously unselected package python3-setuptools-whl.
Preparing to unpack .../python3-setuptools-whl_68.1.2-2ubuntu1.2_all.deb ...
```

Making necessary installations

```
sistla_keerthana@Sistla: ~/kai

sistla_keerthana@Sistla:~/kafka_project$ python3 -m venv venv
sistla_keerthana@Sistla:~/kafka_project$ source venv/bin/activate
(venv) sistla_keerthana@Sistla:~/kafka_project$ pip install kafka-python pymongo requests pyspark
Collecting kafka-python
  Downloading kafka_python-2.3.0-py2.py3-none-any.whl.metadata (10.0 kB)
Collecting pymongo
  Downloading pymongo-4.16.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (10.0 kB)
Collecting requests
  Downloading requests-2.33.1-py3-none-any.whl.metadata (4.8 kB)
Collecting pyspark
  Downloading pyspark-4.1.1.tar.gz (455.4 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 455.4/455.4 MB 1.2 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Collecting dnspython<3.0.0,>=2.6.1 (from pymongo)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Collecting charset_normalizer<4,>=2 (from requests)
  Downloading charset_normalizer-3.4.7-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (40
kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 40.9/40.9 kB 2.5 MB/s eta 0:00:00
Collecting idna<4,>=2.5 (from requests)
  Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Collecting urllib3<3,>=1.26 (from requests)
  Downloading urllib3-2.6.3-py3-none-any.whl.metadata (6.9 kB)
Collecting certifi>=2023.5.7 (from requests)
  Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Collecting py4j<0.10.9.10,>=0.10.9.7 (from pyspark)
  Downloading py4j-0.10.9.9-py2.py3-none-any.whl.metadata (1.3 kB)
Downloading kafka_python-2.3.0-py2.py3-none-any.whl (326 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 326.3/326.3 kB 8.8 MB/s eta 0:00:00
Downloading pymongo-4.16.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (1.7 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.7/1.7 MB 10.6 MB/s eta 0:00:00
Downloading requests-2.33.1-py3-none-any.whl (64 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 64.9/64.9 kB 2.1 MB/s eta 0:00:00
Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 153.7/153.7 kB 8.4 MB/s eta 0:00:00
```

IMPORTANT RULE

Every time you open a new terminal, you must run:

source kafka-env/bin/activate

Otherwise Python won’t find the package.

Using Virtual Environment

```
sistla_keerthana@Sistla: ~/kai
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ bin/kafka-topics.sh --describe \
--topic stock-data \
--bootstrap-server localhost:9092
Topic: stock-data       TopicId: 9j-fF1nTQ52ZBAP80kPvFg PartitionCount: 3       ReplicationFactor: 1    Configs:
        Topic: stock-data       Partition: 0    Leader: 0       Replicas: 0     Isr: 0
        Topic: stock-data       Partition: 1    Leader: 0       Replicas: 0     Isr: 0
        Topic: stock-data       Partition: 2    Leader: 0       Replicas: 0     Isr: 0
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 -m venv kafka-env
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ source kafka-env/bin/activate
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ pip install kafka-python
Collecting kafka-python
  Using cached kafka_python-2.3.0-py2.py3-none-any.whl.metadata (10.0 kB)
Using cached kafka_python-2.3.0-py2.py3-none-any.whl (326 kB)
Installing collected packages: kafka-python
Successfully installed kafka-python-2.3.0
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ pip list
Package      Version
------------ -------
kafka-python 2.3.0
pip          24.0
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$
```

Producer - In Terminal

```
sistla_keerthana@Sistla: ~/kai
sistla_keerthana@Sistla:~$ cd ~/kafka_2.13-3.7.0
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ bin/kafka-console-producer.sh \
--topic stock-data \
--bootstrap-server localhost:9092
>AAPL,250
TSLA,300
INFY,200>>
sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ cd ~/kafka_2.13-3.7.0
```

Consumer - In Terminal

```
sistla_keerthana@Sistla: ~/kai
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano consumer_ops.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consumer_ops.py

Original: AAPL,250
Upper: AAPL
Add 10: 260.0
Double: 500.0
Status: LOW

Original: TSLA,300
Upper: TSLA
Add 10: 310.0
Double: 600.0
Status: HIGH

Original: GOOG,280
Upper: GOOG
Add 10: 290.0
Double: 560.0
Status: HIGH
```

Using auto producer

sistla_keerthana@Sistla: ~/kaf

GNU nano 7.2 producer_api.py

```python
from kafka import KafkaProducer
import json
import time
import random

producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode()
)

symbols = ["AAPL", "TSLA", "INFY", "GOOG", "MSFT"]

i = 0

while True:
    symbol = random.choice(symbols)
    price = random.randint(200, 350)

    data = {
        "symbol": symbol,
        "price": price,
        "seq": i
    }

    partition = i % 3

    producer.send("stock-data", value=data, partition=partition)

    print(f"Sent: {data} → Partition: {partition}")

    i += 1
    time.sleep(2)
```

sistla_keerthana@Sistla: ~/kaf

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano producer_api.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 producer_api.py
Sent: {'symbol': 'AAPL', 'price': 241, 'seq': 0} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 267, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 248, 'seq': 2} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 317, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 310, 'seq': 4} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 278, 'seq': 5} → Partition: 2
Sent: {'symbol': 'INFY', 'price': 238, 'seq': 6} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 346, 'seq': 7} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 333, 'seq': 8} → Partition: 2
Sent: {'symbol': 'INFY', 'price': 245, 'seq': 9} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 211, 'seq': 10} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 322, 'seq': 11} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 235, 'seq': 12} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 216, 'seq': 13} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 297, 'seq': 14} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 257, 'seq': 15} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 239, 'seq': 16} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 232, 'seq': 17} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 281, 'seq': 18} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 201, 'seq': 19} → Partition: 1
Sent: {'symbol': 'INFY', 'price': 233, 'seq': 20} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 333, 'seq': 21} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 256, 'seq': 22} → Partition: 1
Sent: {'symbol': 'INFY', 'price': 213, 'seq': 23} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 292, 'seq': 24} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 286, 'seq': 25} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 277, 'seq': 26} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 252, 'seq': 27} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 304, 'seq': 28} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 340, 'seq': 29} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 287, 'seq': 30} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 340, 'seq': 31} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 321, 'seq': 32} → Partition: 2
```

Consumer that adds 10 to price of each input

```
sistla_keerthana@Sistla: ~/kaf
GNU nano 7.2                                    consumer_add.py
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest'
)

print("Starting ADD Consumer...\n")

for msg in consumer:
    try:
        data = json.loads(msg.value.decode())

        symbol = data["symbol"]
        price = data["price"]

        result = price + 10

        print(f"ADD → {symbol}: {price} + 10 = {result}")

    except Exception as e:
        print("Error:", e)
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano consumer_add.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consumer_add.py
Starting ADD Consumer...

ADD → AAPL: 300 + 10 = 310
ADD → AAPL: 293 + 10 = 303
ADD → INFY: 217 + 10 = 227
ADD → GOOG: 350 + 10 = 360
ADD → AAPL: 340 + 10 = 350
ADD → TSLA: 336 + 10 = 346
ADD → GOOG: 257 + 10 = 267
ADD → GOOG: 295 + 10 = 305
ADD → INFY: 208 + 10 = 218
ADD → TSLA: 329 + 10 = 339
ADD → INFY: 264 + 10 = 274
ADD → INFY: 271 + 10 = 281
ADD → MSFT: 217 + 10 = 227
ADD → GOOG: 347 + 10 = 357
ADD → AAPL: 226 + 10 = 236
ADD → INFY: 304 + 10 = 314
ADD → MSFT: 211 + 10 = 221
ADD → AAPL: 315 + 10 = 325
ADD → AAPL: 348 + 10 = 358
ADD → MSFT: 320 + 10 = 330
ADD → INFY: 350 + 10 = 360
ADD → INFY: 321 + 10 = 331
ADD → AAPL: 308 + 10 = 318
ADD → TSLA: 309 + 10 = 319
ADD → AAPL: 233 + 10 = 243
ADD → INFY: 236 + 10 = 246
ADD → TSLA: 311 + 10 = 321
ADD → MSFT: 276 + 10 = 286
ADD → MSFT: 226 + 10 = 236
ADD → AAPL: 317 + 10 = 327
ADD → GOOG: 281 + 10 = 291
ADD → INFY: 317 + 10 = 327
```

Consumer that subtracts 10 from price of each input

```
consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest'
)

print("Starting Subtract Consumer...\n")

for msg in consumer:
    try:
        data = json.loads(msg.value.decode())

        symbol = data["symbol"]
        price = data["price"]

        result = price - 10

        print(f"Subtract → {symbol}: {price} - 10 = {result}")
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consumer_subtract.py
Starting Subtract Consumer...

Subtract → TSLA: 223 - 10 = 213
Subtract → AAPL: 303 - 10 = 293
Subtract → GOOG: 209 - 10 = 199
Subtract → GOOG: 270 - 10 = 260
Subtract → MSFT: 341 - 10 = 331
Subtract → MSFT: 294 - 10 = 284
Subtract → INFY: 246 - 10 = 236
Subtract → AAPL: 205 - 10 = 195
Subtract → INFY: 232 - 10 = 222
```

Multiply Price by 2

```
for msg in consumer:
    try:
        data = json.loads(msg.value.decode())
        symbol = data["symbol"]
        price = data["price"]

        print(f"MULTIPLY → {symbol}: {price} * 2 = {price * 2}")
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3
producer_api.py
Starting Producer...

Sent: {'symbol': 'AAPL', 'price': 252, 'seq': 0} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 264, 'seq': 1} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 285, 'seq': 2} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 221, 'seq': 3} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 269, 'seq': 4} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 337, 'seq': 5} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 273, 'seq': 6} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 271, 'seq': 7} → Partition: 1
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consu
mer_multiply.py
MULTIPLY Consumer Started...

MULTIPLY → AAPL: 252 * 2 = 504
MULTIPLY → AAPL: 264 * 2 = 528
MULTIPLY → AAPL: 285 * 2 = 570
MULTIPLY → TSLA: 221 * 2 = 442
MULTIPLY → GOOG: 269 * 2 = 538
MULTIPLY → GOOG: 337 * 2 = 674
MULTIPLY → GOOG: 273 * 2 = 546
MULTIPLY → INFY: 271 * 2 = 542
```

Divide Price by 2

```
print("Divide Consumer Started...\n")

for msg in consumer:
    try:
        data = json.loads(msg.value.decode())
        symbol = data["symbol"]
        price = data["price"]

        print(f"MULTIPLY → {symbol}: {price} / 2 = {price / 2}")

    except Exception as e:
        print("Error:", e)
```

```
sistla_keerthana@Sistla: ~/kaf

(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consumer_divide.py
Divide Consumer Started...

MULTIPLY → AAPL: 200 / 2 = 100.0
MULTIPLY → TSLA: 223 / 2 = 111.5
MULTIPLY → GOOG: 329 / 2 = 164.5
MULTIPLY → GOOG: 320 / 2 = 160.0
MULTIPLY → INFY: 247 / 2 = 123.5
MULTIPLY → MSFT: 278 / 2 = 139.0
MULTIPLY → AAPL: 241 / 2 = 120.5
MULTIPLY → GOOG: 306 / 2 = 153.0
MULTIPLY → TSLA: 328 / 2 = 164.0
MULTIPLY → AAPL: 216 / 2 = 108.0
MULTIPLY → INFY: 299 / 2 = 149.5
MULTIPLY → TSLA: 308 / 2 = 154.0
MULTIPLY → TSLA: 256 / 2 = 128.0
MULTIPLY → MSFT: 273 / 2 = 136.5
MULTIPLY → MSFT: 339 / 2 = 169.5
MULTIPLY → INFY: 237 / 2 = 118.5
MULTIPLY → INFY: 325 / 2 = 162.5
MULTIPLY → TSLA: 340 / 2 = 170.0
```

Data Status

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                                    consumer_status.py
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest'
)

print("STATUS Consumer Started...\n")

for msg in consumer:
    try:
        data = json.loads(msg.value.decode())
        symbol = data["symbol"]
        price = data["price"]

        status = "HIGH" if price > 250 else "LOW"

        print(f"STATUS → {symbol}: {price} = {status}")

    except Exception as e:
        print("Error:", e)
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3
producer_api.py
Starting Producer...

Sent: {'symbol': 'AAPL', 'price': 233, 'seq': 0} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 220, 'seq': 1} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 230, 'seq': 2} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 294, 'seq': 3} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 312, 'seq': 4} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 285, 'seq': 5} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 251, 'seq': 6} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 258, 'seq': 7} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 340, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 312, 'seq': 9} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 241, 'seq': 10} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 323, 'seq': 11} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 257, 'seq': 12} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 231, 'seq': 13} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 212, 'seq': 14} → Partition: 2
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano consumer
_status.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consu
mer_status.py
STATUS Consumer Started...

STATUS → GOOG: 230 = LOW
STATUS → MSFT: 294 = HIGH
STATUS → INFY: 312 = HIGH
STATUS → TSLA: 285 = HIGH
STATUS → GOOG: 251 = HIGH
STATUS → GOOG: 258 = HIGH
STATUS → AAPL: 340 = HIGH
STATUS → AAPL: 312 = HIGH
STATUS → AAPL: 241 = LOW
STATUS → TSLA: 323 = HIGH
STATUS → GOOG: 257 = HIGH
STATUS → AAPL: 231 = LOW
STATUS → MSFT: 212 = LOW
STATUS → MSFT: 241 = LOW
STATUS → GOOG: 297 = HIGH
```

Using Mod Operation

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                                    consumer_mod.py
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest'
)

print("MOD Consumer Started...\n")

for msg in consumer:
    try:
        data = json.loads(msg.value.decode())
        symbol = data["symbol"]
        price = data["price"]

        print(f"MOD → {symbol}: {price} % 10 = {price % 10}")

    except Exception as e:
        print("Error:", e)
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3
producer_api.py
Starting Producer...

Sent: {'symbol': 'INFY', 'price': 342, 'seq': 0} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 317, 'seq': 1} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 252, 'seq': 2} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 301, 'seq': 3} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 293, 'seq': 4} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 313, 'seq': 5} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 276, 'seq': 6} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 336, 'seq': 7} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 213, 'seq': 8} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 347, 'seq': 9} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 213, 'seq': 10} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 215, 'seq': 11} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 240, 'seq': 12} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 260, 'seq': 13} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 343, 'seq': 14} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 266, 'seq': 15} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 338, 'seq': 16} → Partition: 1
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano consumer
_mod.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consu
mer_mod.py
MOD Consumer Started...

MOD → GOOG: 252 % 10 = 2
MOD → AAPL: 301 % 10 = 1
MOD → GOOG: 293 % 10 = 3
MOD → GOOG: 313 % 10 = 3
MOD → AAPL: 276 % 10 = 6
MOD → AAPL: 336 % 10 = 6
MOD → AAPL: 213 % 10 = 3
MOD → TSLA: 347 % 10 = 7
MOD → INFY: 213 % 10 = 3
MOD → GOOG: 215 % 10 = 5
MOD → TSLA: 240 % 10 = 0
MOD → MSFT: 260 % 10 = 0
MOD → GOOG: 343 % 10 = 3
MOD → TSLA: 266 % 10 = 6
MOD → AAPL: 338 % 10 = 8
MOD → GOOG: 325 % 10 = 5
MOD → INFY: 228 % 10 = 8
```

Converting to lower case

```python
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest'
)

print("Lower Consumer Started...\n")

for msg in consumer:
    try:
        data = json.loads(msg.value.decode())
        symbol = data["symbol"]

        print(f"SYMBOL → {symbol} → {symbol.lower()}")

    except Exception as e:
        print("Error:", e)
```

```
sistla_keerthana@Sistla: ~/kaf

Starting Producer...

Sent: {'symbol': 'AAPL', 'price': 269, 'seq': 0} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 200, 'seq': 1} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 229, 'seq': 2} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 289, 'seq': 3} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 312, 'seq': 4} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 247, 'seq': 5} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 263, 'seq': 6} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 327, 'seq': 7} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 213, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 345, 'seq': 9} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 222, 'seq': 10} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 345, 'seq': 11} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 334, 'seq': 12} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 315, 'seq': 13} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 272, 'seq': 14} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 279, 'seq': 15} → Partition: 0
Sent: {'symbol': 'INFY', 'price': 225, 'seq': 16} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 204, 'seq': 17} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 271, 'seq': 18} → Partition: 0
```

```
sistla_keerthana@Sistla: ~/kaf

Lower Consumer Started...

SYMBOL → AAPL → aapl
SYMBOL → AAPL → aapl
SYMBOL → AAPL → aapl
SYMBOL → TSLA → tsla
SYMBOL → GOOG → goog
SYMBOL → MSFT → msft
SYMBOL → AAPL → aapl
SYMBOL → INFY → infy
SYMBOL → TSLA → tsla
SYMBOL → AAPL → aapl
SYMBOL → INFY → infy
SYMBOL → AAPL → aapl
SYMBOL → AAPL → aapl
SYMBOL → AAPL → aapl
SYMBOL → TSLA → tsla
SYMBOL → TSLA → tsla
SYMBOL → INFY → infy
SYMBOL → TSLA → tsla
SYMBOL → MSFT → msft
SYMBOL → INFY → infy
SYMBOL → MSFT → msft
```

Converting to upper case

```python
        data = json.loads(msg.value.decode())
        symbol = data["symbol"]

        print(f"SYMBOL → {symbol} → {symbol.upper()}")
```

Advanced Consumer - Calculating everything for each input

```
sistla_keerthana@Sistla: ~/kaf

(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano consumer_advanced.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 consumer_advanced.py

============================
DATA: {'symbol': 'AAPL', 'price': 241, 'seq': 0}
Add: 251
Sub: 236
Mul: 482
Div: 120.5
Mod: 1
Upper: AAPL
Lower: aapl
Length: 4
Round: 241
Abs: 241
Status: LOW
Partition: 0
Offset: 0

============================
DATA: {'symbol': 'AAPL', 'price': 317, 'seq': 3}
Add: 327
Sub: 312
Mul: 634
Div: 158.5
Mod: 7
Upper: AAPL
Lower: aapl
Length: 4
Round: 317
Abs: 317
Status: HIGH
Partition: 0
Offset: 1

============================
DATA: {'symbol': 'INFY', 'price': 238, 'seq': 6}
```

Importing Data Via API (from FINNHUB)

```
sistla_keerthana@Sistla: ~/kaf

  GNU nano 7.2                                                    producer_api.py
producer = KafkaProducer(
    bootstrap_servers='localhost:9092',
    value_serializer=lambda v: json.dumps(v).encode()
)

symbols = ["AAPL", "TSLA", "MSFT", "GOOG", "AMZN"]

i = 0

print("Starting Finnhub Producer...\n")

while True:
    for symbol in symbols:
        try:
            url = f"https://finnhub.io/api/v1/quote?symbol={symbol}&token={API_KEY}"
            response = requests.get(url)
            data = response.json()

            price = data.get("c")

            if price is None or price == 0:
                print("nvalid data:", data)
                continue

            message = {
                "symbol": symbol,
                "price": price,
                "seq": i
            }

            partition = i % 3

            producer.send("stock-data", value=message, partition=partition)
```

MOVING AVERAGE (last 5 values)

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano produ
cer_api.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 pr
oducer_api.py
Starting Finnhub Producer...

Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 0} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 2} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 4} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 5} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 6} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 7} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 9} → Partition: 0
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano co
nsumer_upper.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano co
nsumer_moving_avg.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3
 consumer_moving_avg.py
AAPL MOVING AVG (5): 255.94
TSLA MOVING AVG (5): 308.27
MSFT MOVING AVG (5): 330.01
GOOG MOVING AVG (5): 321.13
AMZN MOVING AVG (5): 298.86
AAPL MOVING AVG (5): 298.86
TSLA MOVING AVG (5): 298.86
MSFT MOVING AVG (5): 298.86
GOOG MOVING AVG (5): 298.86
AMZN MOVING AVG (5): 298.86
```

VOLATILITY (price change)

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                        consumer_volatility.py *
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest',
    value_deserializer=lambda x: json.loads(x.decode())
)

last_price = None

for msg in consumer:
    price = msg.value["price"]
    symbol = msg.value["symbol"]

    if last_price is not None:
        change = price - last_price
        print(f"{symbol} CHANGE:", round(change, 2))

    last_price = price
```

```
sistla_keerthana@Sistla: ~/kaf
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 pro
ducer_api.py
Starting Finnhub Producer...

Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 0} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 2} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 4} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 5} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 6} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 7} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 9} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 10} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 11} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 12} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 13} → Partition: 1
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 14} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 15} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 16} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 17} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 18} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 19} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 20} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 21} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 22} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 23} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 24} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 25} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 26} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 27} → Partition: 0
```

```
sistla_keerthana@Sistla: ~/kaf
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano c
onsumer_volatility.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python
3 consumer_volatility.py
TSLA CHANGE: 104.67
MSFT CHANGE: 12.87
GOOG CHANGE: -79.0
AMZN CHANGE: -84.69
AAPL CHANGE: 46.15
TSLA CHANGE: 104.67
MSFT CHANGE: 12.87
GOOG CHANGE: -79.0
AMZN CHANGE: -84.69
AAPL CHANGE: 46.15
TSLA CHANGE: 104.67
MSFT CHANGE: 12.87
GOOG CHANGE: -79.0
AMZN CHANGE: -84.69
AAPL CHANGE: 46.15
TSLA CHANGE: 104.67
MSFT CHANGE: 12.87
GOOG CHANGE: -79.0
AMZN CHANGE: -84.69
AAPL CHANGE: 46.15
TSLA CHANGE: 104.67
MSFT CHANGE: 12.87
GOOG CHANGE: -79.0
AMZN CHANGE: -84.69
AAPL CHANGE: 46.15
TSLA CHANGE: 104.67
MSFT CHANGE: 12.87
GOOG CHANGE: -79.0
```

PRICE TREND (UP/DOWN)

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                        consumer_trend.py *
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest',
    value_deserializer=lambda x: json.loads(x.decode())
)

last_price = None

for msg in consumer:
    price = msg.value["price"]
    symbol = msg.value["symbol"]

    if last_price is not None:
        if price > last_price:
            print(f"{symbol} TREND: UP 📈")
        elif price < last_price:
            print(f"{symbol} TREND: DOWN 📉")
        else:
            print(f"{symbol} TREND: SAME")

    last_price = price
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 pro
ducer_api.py
Starting Finnhub Producer...

Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 0} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 2} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 4} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 5} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 6} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 7} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 9} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 10} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 11} → Partition: 2
```

```
onsumer_volatility.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano c
onsumer_trend.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python
3 consumer_trend.py
TSLA TREND: UP 📈
MSFT TREND: UP 📈
GOOG TREND: DOWN 📉
AMZN TREND: DOWN 📉
AAPL TREND: UP 📈
TSLA TREND: UP 📈
MSFT TREND: UP 📈
GOOG TREND: DOWN 📉
AMZN TREND: DOWN 📉
AAPL TREND: UP 📈
TSLA TREND: UP 📈
MSFT TREND: UP 📈
```

SPIKE DETECTION (>20 jump)

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                    consumer_spike.py *
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest',
    value_deserializer=lambda x: json.loads(x.decode())
)

last_price = None

for msg in consumer:
    price = msg.value["price"]
    symbol = msg.value["symbol"]

    if last_price is not None and abs(price - last_price) > 20:
        print(f"🚨 SPIKE DETECTED in {symbol}: {price}")

    last_price = price
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 pro
ducer_api.py
Starting Finnhub Producer...

Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 0} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 2} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 4} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 5} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 6} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 7} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 9} → Partition: 0
```

```
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano c
onsumer_spike.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python
3 consumer_spike.py
🚨 SPIKE DETECTED in TSLA: 360.61
🚨 SPIKE DETECTED in GOOG: 294.48
🚨 SPIKE DETECTED in AMZN: 209.79
🚨 SPIKE DETECTED in AAPL: 255.94
🚨 SPIKE DETECTED in TSLA: 360.61
🚨 SPIKE DETECTED in GOOG: 294.48
🚨 SPIKE DETECTED in AMZN: 209.79
🚨 SPIKE DETECTED in AAPL: 255.94
🚨 SPIKE DETECTED in TSLA: 360.61
```

MIN / MAX TRACKER

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                    consumer_min_max.py
from kafka import KafkaConsumer
import json

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest',
    value_deserializer=lambda x: json.loads(x.decode())
)

min_price = float('inf')
max_price = float('-inf')

for msg in consumer:
    price = msg.value["price"]
    symbol = msg.value["symbol"]

    min_price = min(min_price, price)
    max_price = max(max_price, price)

    print(f"{symbol} MIN:", min_price, "MAX:", max_price)
```

```
sistla_keerthana@Sistla: ~/kaf
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 pro
ducer_api.py
Starting Finnhub Producer...

Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 0} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 2} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 4} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 5} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 6} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 7} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 9} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 10} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 11} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 12} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 13} → Partition: 1
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 14} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 15} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 16} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 17} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 18} → Partition: 0
```

```
sistla_keerthana@Sistla: ~/kaf
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano c
onsumer_min_max.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python
3 consumer_min_max.py
AAPL MIN: 255.94 MAX: 255.94
TSLA MIN: 255.94 MAX: 360.61
MSFT MIN: 255.94 MAX: 373.48
GOOG MIN: 255.94 MAX: 373.48
AMZN MIN: 209.79 MAX: 373.48
AAPL MIN: 209.79 MAX: 373.48
TSLA MIN: 209.79 MAX: 373.48
MSFT MIN: 209.79 MAX: 373.48
GOOG MIN: 209.79 MAX: 373.48
AMZN MIN: 209.79 MAX: 373.48
AAPL MIN: 209.79 MAX: 373.48
TSLA MIN: 209.79 MAX: 373.48
MSFT MIN: 209.79 MAX: 373.48
GOOG MIN: 209.79 MAX: 373.48
AMZN MIN: 209.79 MAX: 373.48
AAPL MIN: 209.79 MAX: 373.48
TSLA MIN: 209.79 MAX: 373.48
MSFT MIN: 209.79 MAX: 373.48
GOOG MIN: 209.79 MAX: 373.48
```

Z-SCORE (Anomaly detection)

```
sistla_keerthana@Sistla: ~/kaf
  GNU nano 7.2                                consumer_zscore.py
from kafka import KafkaConsumer
import json
import statistics

consumer = KafkaConsumer(
    'stock-data',
    bootstrap_servers='localhost:9092',
    auto_offset_reset='latest',
    value_deserializer=lambda x: json.loads(x.decode())
)

prices = []

for msg in consumer:
    price = msg.value["price"]
    symbol = msg.value["symbol"]

    prices.append(price)

    if len(prices) > 5:
        mean = statistics.mean(prices)
        std = statistics.stdev(prices)

        if std != 0:
            z = (price - mean) / std
            print(f"{symbol} Z-SCORE:", round(z, 2))
```

```
sistla_keerthana@Sistla: ~/kaf
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3 pro
ducer_api.py
Starting Finnhub Producer...

Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 0} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 1} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 2} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 3} → Partition: 0
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 4} → Partition: 1
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 5} → Partition: 2
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 6} → Partition: 0
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 7} → Partition: 1
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 8} → Partition: 2
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 9} → Partition: 0
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 10} → Partition: 1
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 11} → Partition: 2
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 12} → Partition: 0
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 13} → Partition: 1
Sent: {'symbol': 'AMZN', 'price': 209.79, 'seq': 14} → Partition: 2
Sent: {'symbol': 'AAPL', 'price': 255.94, 'seq': 15} → Partition: 0
Sent: {'symbol': 'TSLA', 'price': 360.61, 'seq': 16} → Partition: 1
Sent: {'symbol': 'MSFT', 'price': 373.48, 'seq': 17} → Partition: 2
Sent: {'symbol': 'GOOG', 'price': 294.48, 'seq': 18} → Partition: 0
```

```
sistla_keerthana@Sistla: ~/kaf
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ nano co
nsumer_zscore.py
(kafka-env) sistla_keerthana@Sistla:~/kafka_2.13-3.7.0$ python3
 consumer_zscore.py
AAPL Z-SCORE: -0.56
TSLA Z-SCORE: 0.92
MSFT Z-SCORE: 0.97
GOOG Z-SCORE: -0.23
AMZN Z-SCORE: -1.36
AAPL Z-SCORE: -0.62
TSLA Z-SCORE: 0.95
MSFT Z-SCORE: 1.06
GOOG Z-SCORE: -0.17
AMZN Z-SCORE: -1.39
AAPL Z-SCORE: -0.64
TSLA Z-SCORE: 0.97
MSFT Z-SCORE: 1.1
GOOG Z-SCORE: -0.15
AMZN Z-SCORE: -1.4
AAPL Z-SCORE: -0.65
TSLA Z-SCORE: 0.97
MSFT Z-SCORE: 1.12
GOOG Z-SCORE: -0.13
```